

Visualizing the Moon for Artemis II

Ernie Wright
Scientific Visualization Studio
NASA Goddard Space Flight Center
USA
STC
USA
ernest.t.wright@nasa.gov

Kel Elkins
Scientific Visualization Studio
NASA Goddard Space Flight Center
USA
STC
USA
kelly.elkins@nasa.gov

Marie Henderson
Sciences and Exploration Directorate
NASA Goddard Space Flight Center
USA
University of Maryland Baltimore
County
USA
marie.j.henderson@nasa.gov



Figure 1: Visualizing the Moon for Artemis II

ACM Reference Format:

Ernie Wright, Kel Elkins, and Marie Henderson. 2026. Visualizing the Moon for Artemis II. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference (SIGGRAPH Production Sessions '26)*, July 19–23, 2026,

Los Angeles, CA, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3774846.3797550>

1 Overview

Artemis II is the first crewed spaceflight to the Moon, in fact the first beyond Earth orbit, since Apollo 17 in 1972. The four astronauts will fly around the far side of the Moon on the sixth day of their mission, and for a six-hour period during the flyby, they will be observing, photographing, and describing the geology of the side of the Moon never seen from Earth, including areas never seen by human eyes in daylight even by Apollo astronauts.



This work is licensed under a Creative Commons Attribution 4.0 International License. *SIGGRAPH Production Sessions '26*, Los Angeles, CA, USA
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2297-4/2026/07
<https://doi.org/10.1145/3774846.3797550>

Cinematic scientific visualization using high-end commercial 3D rendering software has played a crucial role in both planning the observations and in training the astronauts. This kind of visualization combines rendering realism with large NASA datasets to create highly accurate representations of the lighting conditions the astronauts are likely to encounter.

Small differences in Sun angle can reveal or hide topographic, albedo, and color features of the greatest interest to lunar scientists. For this reason, the final observation plan won't be created until after the launch, when the precise trajectory and lunar arrival time are known. A six-hour realtime visualization of the flyby will be rendered from Mission Control on day 2 of the flight. This will be used by the science team to create the observation plan, which will then be uploaded to the crew as they approach the Moon.

The visualization will later be played on a large screen in Mission Control, synchronized with the actual flyby.

The production session will exhibit several visualizations of the Artemis II lunar flyby and explain how they were created and how they were used by lunar scientists and the astronauts. The visualizations draw on publicly available NASA datasets and spaceflight software libraries that will be highlighted and explored.

As of this writing, the 10-day Artemis II mission is scheduled to launch in early March. If it is completed successfully before SIGGRAPH, we will compare the flyby visualizations to photographs and video taken during the mission, and we will hear from the astronauts how useful the visualizations were to them during the flight.

The availability of panelists may be contingent on the success of the mission and on government funding and travel approval.